

AutoGuardian – Hardware Component List

Components
ESP32 DevKitC (ESP32-WROOM-32)
SIM800L GSM Module (SIM800L or SIM808)
GPS Module NEO-6M or NEO-M8N
Automotive-rated Fuel Kill Relay (30-40A SPST)
Relay Socket + 5-pin base
Logic-level MOSFET (e.g. IRLZ44N) or NPN transistor
Flyback Diode (1N4007)
TVS Diode for 12V line (e.g. SMBJ58A)
Automotive-rated DC-DC Buck Converter (12V-5V)
Voltage Regulator for SIM800L (4.0V)
Inline Fuse Holders + Fuses (2-5A, 20-30A)
Automotive Wiring Kit
Crimp Terminals, Connectors and Screw Terminals
Manual Override SPST Switch (sealed)
Active GPS Antenna (external)
Buzzer (active)
TTP223 Touch Sensor / Tamper Sensor
MPU-6050 Gyroscope/Accelerometer
Automotive Enclosure (IP54/IP65)
Perma-Proto Board / Automotive PCB
Heat Shrink Tubing, Cable Ties, Wire Loom
Soldering Consumables
2-Channel Relay Module (optically isolated)
OLED Display (SSD1306 0.96" 12C)
CAN-Bus OBD-II Interface
Thermal & Vibration Mounts
Multimeter + Test Leads